

ShopEZ E-Commerce Application Documentation

1. Project Overview

Project Name: ShopEZ – E-Commerce Application

ShopEZ is a full-stack e-commerce web application developed using **React, TypeScript, Vite, Express.js, and Tailwind CSS**. The application provides a complete online shopping experience for customers and a powerful administration panel for managing products, users, and orders.

The system supports two user roles:

- **Customer**
- **Administrator**

Both users have separate dashboards and permissions.

2. Project Objectives

The primary objectives of ShopEZ are:

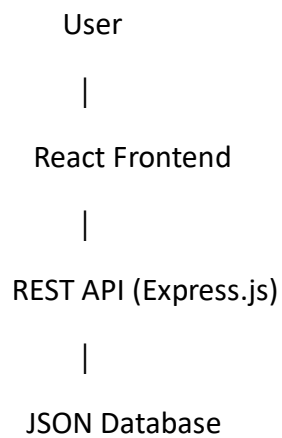
- Build a modern online shopping platform.
 - Allow customers to browse and purchase products.
 - Provide secure authentication.
 - Enable administrators to manage products and orders.
 - Maintain user-friendly and responsive UI.
 - Demonstrate full-stack development using React and Express.
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3. Technology Stack

Technology	Purpose
React 19	Frontend UI
TypeScript	Type Safety
Vite	Build Tool

Technology	Purpose
Express.js	Backend Server
Tailwind CSS v4	Styling
Node.js	Runtime Environment
JSON Database	Data Storage
Local Storage	Session Management

4. System Architecture



The application follows a **Client–Server Architecture**.

- Frontend communicates with backend using REST APIs.
 - Backend processes requests.
 - JSON database stores users, products, and orders.
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5. Project Directory Structure

ShopEZ

```
|
|
|— server.ts
|— package.json
```

```
└─ tsconfig.json
└─ vite.config.ts
└─ metadata.json
└─ .env.example
|
└─ server
|   └─ db.ts
|   └─ routes.ts
|
└─ src
    └─ main.tsx
    └─ App.tsx
    └─ index.css
    └─ types.ts
    |
    └─ components
    |   └─ Navbar.tsx
    |   └─ ProductCard.tsx
    |
    └─ pages
        └─ LandingPage.tsx
        └─ ProductsPage.tsx
        └─ ProductDetailPage.tsx
        └─ AuthPage.tsx
        └─ UserProfile.tsx
```

└─ CartPage.tsx

└─ AdminDashboard.tsx

└─ AdminProducts.tsx

└─ AdminNewProduct.tsx

└─ AdminUsers.tsx

└─ AdminOrders.tsx

6. Module Description

6.1 Authentication Module

Features:

- User Registration
- User Login
- Session Management
- Local Storage Authentication
- Role-based Access

Roles:

- Customer
 - Admin
-

6.2 Landing Page

Features:

- Promotional Banner
- Product Categories
- Featured Products
- Responsive Design
- Dynamic Content

6.3 Product Module

Features:

- Product Listing
- Search Products
- Category Filter
- Price Sorting
- Discount Sorting
- Product Cards
- Product Details

6.4 Product Details Module

Features:

- Multiple Images
- Product Carousel
- Size Selection
- Stock Checking
- Dynamic Pricing

6.5 Shopping Cart

Features:

- Add to Cart
- Remove Product
- Quantity Management
- Total Price Calculation
- Discount Calculation

6.6 Checkout Module

Customer enters:

- Full Name
- Mobile Number
- Shipping Address
- Pincode
- Payment Method

Payment Options:

- Cash on Delivery
- Card Payment
- Net Banking

6.7 User Profile

Features:

- Personal Information
- Previous Orders
- Order History
- Order Status
- Cancel Order

Order Status:

- Order Placed
 - In Transit
 - Delivered
 - Cancelled
-

7. Admin Module

The Admin Dashboard provides complete management of the application.

7.1 Dashboard

Displays:

- Total Users
 - Total Products
 - Active Orders
 - Live Statistics
-

7.2 Product Management

Admin can:

- Add Products
 - Edit Products
 - Delete Products
 - Update Prices
 - Upload Image URL
 - Manage Categories
 - Manage Sizes
-

7.3 User Management

Admin can:

- View Registered Users
- View Email IDs
- Check User Roles
- Manage Customer Records

7.4 Order Management

Admin can:

- View Orders
 - Update Order Status
 - Track Delivery
 - Manage Dispatch
 - Monitor Shipping Information
-

8. Database Models

User

Stores:

- User ID
- Name
- Email
- Password
- Role

Roles:

- Admin
 - Customer
-

Product

Stores:

- Product ID
- Product Name
- Description

- Price
 - Discount
 - Category
 - Sizes
 - Image URL
-

CartItem

Stores:

- Product
 - Quantity
 - Selected Size
 - Total Price
-

Order

Stores:

- Order ID
- Customer Details
- Shipping Address
- Payment Method
- Products
- Order Date
- Status

Statuses:

- Order Placed
- In Transit
- Delivered

- Cancelled
-

9. Key Features

Customer Features

- User Registration
 - Login
 - Browse Products
 - Product Search
 - Category Filter
 - Price Sorting
 - Product Details
 - Add to Cart
 - Checkout
 - Order Tracking
 - Profile Management
 - Order History
-

Admin Features

- Admin Login
 - Dashboard
 - Product Management
 - User Management
 - Order Management
 - Banner Management
 - Inventory Management
-

10. Session Management

The application stores login information using **Local Storage**.

Benefits:

- Persistent Login
 - Fast Authentication
 - Role Verification
 - Secure Session Handling
-

11. REST API

The backend provides REST APIs for:

Authentication

- Login
- Register
- Logout

Products

- Get Products
- Add Product
- Update Product
- Delete Product

Users

- Get Users
- Update User

Orders

- Create Order
- Get Orders
- Update Order Status

12. User Flow

Customer Flow

Register/Login



Landing Page



Browse Products



Product Details



Add to Cart



Checkout



Order Confirmation



Order Tracking

Admin Flow

Admin Login



Dashboard



Manage Products



Manage Users



Manage Orders



Update Banner

13. Installation Guide

Step 1

Clone the repository.

```
git clone <repository-url>
```

Step 2

Open the project in Visual Studio Code.

Step 3

Install Dependencies

```
npm install
```

Step 4

Run Development Server

```
npm run dev
```

Server URL:

```
http://localhost:3000
```

Step 5 (Optional)

Run on Port 8000

Linux / macOS

PORT=8000 npm run dev

Windows PowerShell

\$env:PORT=8000; npm run dev

Windows CMD

set PORT=8000 && npm run dev

Step 6

Production Build

npm run build

npm run start

14. Demo Credentials

Customer Login

Email

customer@gmail.com

Password

customer123

Admin Login

Email

admin@gmail.com

Password

admin123

15. Advantages

- Responsive Design
 - Modern UI
 - Secure Authentication
 - Role-Based Access
 - Easy Product Management
 - Live Order Tracking
 - Fast Performance
 - Simple Deployment
 - Scalable Architecture
 - User-Friendly Interface
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16. Future Enhancements

- Online Payment Gateway Integration
 - Email Notifications
 - Wishlist Feature
 - Product Reviews & Ratings
 - AI Product Recommendations
 - Coupon & Discount System
 - Inventory Analytics Dashboard
 - Multi-Vendor Support
 - Dark Mode
 - Push Notifications
 - Real Database Integration (MongoDB/MySQL)
 - Cloud Deployment (AWS, Azure, Vercel)
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17. Conclusion

ShopEZ is a modern full-stack e-commerce application that demonstrates industry-standard web development practices using React, TypeScript, Express.js, and Tailwind CSS. The project offers a complete shopping experience for customers while providing administrators with efficient tools for managing products, users, and orders. Its modular architecture, responsive design, secure role-based authentication, and scalable structure make it an excellent academic project as well as a strong portfolio project for showcasing full-stack development skills.